Improper integrals

Key Points

An integral is improper its integration range

- Contains undefined points for the integrand

Or

- Is infinity

We deal with the problem of undefined points by splitting up the integral and avoiding these points.

We deal with the problem of an infinite integration range by taking the limit.

Note. Improper integrals may diverge

Basic Skills

Q1. State if and why the following definite integrations are improper

a) $\int_0^\infty e^{-x}\, dx$ b) $\int_0^{10} \frac{1}{x-4}\, dx$

c) $\int_0^5 \frac{x-4}{x+1}\, dx$ d) $\int_{-5}^5 \frac{1}{x^2+5}\, dx$

e) $\int_{-\infty}^\infty e^{x^2}\, dx$ f) $\int_4^{-4} \frac{1}{(x-4)(x+4)}\, dx$

Q2. Explain the difference between the two types of improper integrals and state whether it is always possible to calculate the improper integral

Q3. By using the fact that $\int_2^\infty \frac{1}{x}\, dx = \lim_{k\to\infty} \int_2^k \frac{1}{x}\, dx$, work out the improper integral $\int_2^\infty \frac{1}{x}\, dx$ exists

Competency Skills

Q4. By using the fact that $\int_4^{10} \frac{1}{\sqrt{x-4}}\, dx = \lim_{k\to 4} \int_k^{10} \frac{1}{\sqrt{x-4}} dx$ work out if the value exists and if so find it

Q5. Work out the following improper integrals, if they exist

a) $\int_2^\infty \frac{1}{x^2}\, dx$ b) $\int_0^1 \frac{1}{\sqrt{x}}\, dx$

c) $\int_1^\infty \frac{3}{\sqrt[3]{x}}\, dx$ d) $\int_2^4 \frac{1}{(x-3)^2}\, dx$

Q6. Given that $\lim_{x\to 0} x \ln x = 0$ determine if the improper integral $\int_0^e \ln x\, dx$ exists

Q7. Determine if the improper integral $\int_0^\infty \frac{e^{-x}}{1+e^{-x}}\, dx$ exists

Q8. By using partial fractions show that the improper integral $\int_1^4 \frac{1}{(x-2)(x-3)}\, dx$ diverges

Advanced Skills

Q9. Work out the improper integral $\int_0^\infty xe^{-x}\, dx$ by using l'hospital's rule as needed

Q10. Decide which are these two integrals exists, $\int_1^\infty \frac{1}{\sqrt{x^3}}\, dx$, $\int_1^\infty \frac{1}{\sqrt[3]{x^2}}\, dx$

Q11. a) In how many ways is the integral $\int_{-\infty}^\infty \frac{1}{x^2}\, dx$ improper

b) Determine if the integral $\int_{-\infty}^\infty \frac{1}{x^2}\, dx$ exists

Q12. Evaluate $\int_0^\infty \frac{1}{x+5} - \frac{1}{x+1}$

Q13. A student argues that "As the graph $y = x$ is odd, meaning that if negative x then y is the negative of when x is positive, then the integral $\int_{-\infty}^\infty x\, dx = 0$" comment on this students thought

Extension

Q14. For an integrand of the form x^p can you decide a rule to do with p for when its improper integral $\int_{-1}^1 x^p\, dx$ diverges

Hint: Look at Q9 for a starting point